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The Fire Model of Trading Why the Human Brain Was Always Going to Be Drawn to Markets



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The Fire Model of Trading

Why the Human Brain Was Always Going to Be Drawn to Markets

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Ancient parables. Modern markets. The gap between the two is smaller than you think.

An Essay in Neuroscience and Trading (not a parable — a deep dive)

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I was sitting by the fire pit last week doing something I genuinely love — splitting logs with the hydraulic splitter, feeding the fire, watching it work. No romantic nonsense about wielding an ax. I'm 72. The hydraulic splitter is a beautiful machine and I have no ego about it.

But sitting there watching the fire, I noticed something. I wasn't just warm. I was engaged. Something in me was paying attention in a way that felt older than habit. More like instinct. The movement of the flames, the unpredictability of it, the satisfaction of managing it — I didn't want to look away.

I'd been reading about how culture and genetics don't just coexist — they shape each other across generations. That certain cultural practices, repeated long enough, actually feed back into human biology. Fire being one of the oldest and most profound examples.

And sitting there by the pit, it hit me the way things hit you when you're not trying to think: there's an analog with trading. Not a loose metaphor. A structural one. The same ancient circuits. The same pull.

I just knew it. So I went and mapped it out



Long before humans mastered fire, certain traits were already baked in. Neural circuits. Behavioral tendencies. Things that evolved for completely

different purposes — survival, social bonding, pattern recognition, risk calibration. None of them evolved for fire. But when fire showed up, those circuits lit up like they'd been waiting their whole evolutionary life for it.

Those same circuits are waiting at your trading terminal right now.

Markets are the modern fire. Unpredictable. Patterned. Dangerous. Rewarding. Socially transmitted. Cognitively rich. The attraction you feel isn't weakness. Isn't gambling pathology. It's an inheritance — millions of years old — finally finding a new hearth.

Here's the architecture.

Novelty-Seeking Dopamine Circuits

Fire was a novelty engine. Flicker. Crackle. Motion. Unpredictable heat. The human dopamine system evolved to approach dynamic, information-rich stimuli. That same circuit fires when price moves. Volatility bursts, news shocks, tape acceleration — your brain treats all of it the same way early humans treated flames.

Markets feel alive because your brain classifies them as alive.

That's not metaphor. That's neuroscience.

Pattern-Recognition Architecture

Flames produce fractal-like patterns. Irregular. Recursive. Never exactly the same twice. Your visual cortex evolved to extract structure from exactly that kind of noise — tracking predators, reading weather, mapping terrain. That same machinery now locks onto candlestick charts, order flow, and volatility clusters.

You're not being fooled by charts. You're doing exactly what four million years of evolution built you to do.

The problem is that markets are also very good at producing false patterns. The machinery doesn't know the difference between a signal and a mirage. That's where experience comes in. That's where structure matters.

That's what this whole framework is about.

Risk-Tolerant Prefrontal Override

Fire triggered fear. Every mammal on the savanna fled from it. Humans didn't. Because the human prefrontal cortex could do something other mammals couldn't — override the amygdala long enough to ask: what if this is useful?

Danger became danger-with-upside. Risk became opportunity.

Trading runs on the same override. The loss risk is real. The amygdala knows it. But the prefrontal cortex reframes it as potential reward and steps toward the flame anyway.

Controlled risk exploration isn't a modern personality trait. It's an ancient adaptation that kept the species alive. You inherited it. So did every other trader you've ever met.

The Poke-at-It Instinct

Human childhood is unusually long compared to other primates. That extended curiosity window — that period of poking at things, asking why, taking things apart — is what made early humans poke at fire instead of fleeing it.

That same instinct is what makes new traders stare at charts for hours. What makes someone want to understand implied volatility at 11 PM when they have work in the morning.

It's not immaturity. It's developmental architecture doing exactly what it was designed to do. Curiosity is a feature. Markets just happen to be a very expensive way to express it without a framework to channel it.

Hyper-Imitation and Social Learning

Fire mastery didn't spread through individual discovery. It spread through watching. Imitating. Transmitting. One group learned to manage fire, and that knowledge moved through social networks the way all dangerous, high-value skills moved — through demonstration, apprenticeship, shared practice.

Sound familiar?

Trading spreads the same way. Mentors. Streams. Annotated charts. Discord threads at 6 AM before the open. That's cumulative culture, not individual genius. The mirror-neuron system that once transmitted fire-handling now transmits market-handling.

This is why community matters in trading. Not for the social experience. For the transmission of hard-won structural knowledge that takes decades to accumulate alone.

The Salience Network

Fire demanded attention. It wasn't optional. The salience network — the system that decides what matters, what gets cognitive priority — treated fire as maximally relevant because it was.

Markets trigger the same system. Volatility spikes. P&L swings. Price at a key level. The screen pulls your attention the way the hearth pulled your ancestor's gaze.

This is why it's so hard to step away from the terminal. It's not addiction in the clinical sense. It's relevance detection working exactly as designed. The brain has tagged markets as important. And in a very real sense — they are.

The discipline isn't about caring less. It's about managing when you look and what you do with what you see.

The Default Mode Network

Here's the one most people don't talk about.

Fire stabilized the mind. The hearth wasn't just warmth and light — it was a cognitive anchor. A place where stories got told, plans got made, meaning organized itself. The brain uses focal stimuli to scaffold thought. Fire was one of the original scaffolds.

Charts do the same thing. For experienced traders, studying a chart is a meditative act. It's not just information extraction. It's the mind using a focal stimulus to think — about structure, about probability, about its own assumptions.

That's not something everyone understands from the outside. But traders know the feeling.

The chart becomes the hearth.

Reward Prediction Error

Dopamine doesn't respond to reward. It responds to the difference between expected and actual reward. The prediction error. The surprise.

Fire delivered constant micro-surprises. Every crackle, every shift in flame, every unexpected gust — tiny deviations from expectation that kept the dopamine system perpetually engaged, perpetually learning.

Every tick in a live market is a prediction error. Every price that doesn't go where structure said it should. Every moment the tape does something you didn't see coming.

This is why markets feel alive even when you're flat. The brain isn't waiting for a trade. It's learning. Constantly recalibrating. That's the dopamine system doing its job.

The danger is mistaking that engagement for edge. Feeling active isn't the same as being right. Forty years of watching this tape will teach you that lesson — one way or another.

Social Cohesion Around Focal Points

The hearth created the tribe. It was the original gathering point — the place where social bonds formed, hierarchies organized, and shared understanding developed around a common focal point.

Markets create communities the same way. Trading floors. Chat rooms. Discord servers. The shared uncertainty of not knowing what the tape will do next creates a kind of social adhesive. You bond with people who are watching the same thing you are, feeling the same tension, navigating the same unknowns.

This is why traders who go it completely alone often struggle — not just for lack of information, but for lack of the social scaffolding that comes from navigating uncertainty in community.

Markets are a modern hearth. Humans have always gathered at the hearth.

The Throughline

Trading isn't a modern quirk. It's an ancient brain encountering a modern fire.

The attraction comes from novelty, pattern, risk, reward, imitation, salience, meaning, and social cohesion — all of it inherited, none of it designed for markets specifically, all of it perfectly suited to them.

These aren't weaknesses to overcome. They're the architecture of the species. The goal isn't to become less human. It's to understand which parts of the inheritance serve you at the terminal — and which parts will burn you if you don't manage them deliberately.

Forty years of watching this tape has taught me one thing above everything else: the traders who last are not the ones who have no emotional response to markets. They're the ones who understand why they have the response they have — and have built a structure that doesn't let that response make their decisions.

That's the work. Not suppression. Understanding.

The flow was always the signal. The brain was always going to respond to it. The question is whether you're going to react to the fire — or learn to use it.

Same question the first guy who didn't run from the flames had to answer. He figured it out. So can you.

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Great metaphor, super insights. Thank's Doc!!



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